

Practice questions on Module 3

1. What is an electric field? Obtain an expression for electric field intensity due to line charge.
2. Obtain an expression for electric field intensity due to surface and volume charge
3. What is Gauss's law for electrostatics? Obtain its integral and differential form.
4. What is electric potential. Obtain relation between electric intensity and electric potential.
5. What is Ampere's law for magnetism? Obtain its integral and differential form.
6. What is Gauss's law for magnetism? Using this prove that magnetic poles do not exist.
monopole.
7. What is Faraday's law of electromagnetism? Obtain its integral and differential form.
8. Why Maxwell modified Ampere's law? Obtain modified Ampere's law. What is displacement current.
9. Write all four Maxwell's equations in differential or integral form.
Explain their significance.
10. Using Maxwell's equations in free space obtain formula for speed of electromagnetic wave.
Prove that light is an electromagnetic wave.
11. What is unpolarised and plane unpolarized light? How is it obtained? Give their representations.
12. Write a note on application of electromagnetic theory (EMT) in Wi-Fi, antennas and bluetooth.